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CPU and GPU Global Memory Quiz

Review Learning Objectives

1. The first CUDA-compatible Nvidia GPU devices had 256 KB of global memory and now has 2 TB of global memory. **1 / 1 point**

- ○ True
- ◉ False

Correct
This is correct because the first CUDA cards had at a minimum 256MB of global memory and now have around 4GB on the high-end.

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Due Aug 30, 11:59 PM IST

2. Which of the following are Nvidia GPU architectures? **1 / 1 point**

- ☐ Radeon
- ☐ Dora
- ☑ Ampere

Correct
This is the architecture that started to be developed in 2020.

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- ☑ Tesla

Correct
This architecture was mostly used from 2008 to 2010.

3. The nvidia-smi CLI tool allows which of the following options and the associated use **1 / 1 point**

- ☑ -i This allows you to query for a specific Nvidia device based on an assigned id

Correct
Correct. An alternative is the --id long form.

- ☑ -L *This lists all NVidia GPUs*

Correct
Correct. you can also use the --list_gpus for similar effect.

- ☑ -x Outputs in XML format

Correct
This is correct, so if you would like to process the output of nvidia-smi as XML this is the option to use.

- ☐ -z Returns the maximum number of zero-byte images that can be loaded onto the card
- ☐ --amd Lists all AMD host devices

4. The lspci command was developed by Nvidia and is installed as part of CUDA expressly for the listing of GPU devices. **1 / 1 point**

- ○ True
- ◉ False

Correct
This tool is used to list all connected PCI devices on any linux machine that it is installed on.

5. Which of the following is an example of a C++ primitive data type? **1 / 1 point**

- ○ const
- ◉ int
- ○ primitive
- ○ std::vector

Correct
This is one of the canonical primitives.

6. Which highlighted (in **bold** text) parts of a pageable memory allocation are not correct? **1 / 1 point**

char *words**::**
input_a = (**char *)cudaMalloc**(1024 * **sizeof**(char));

- ☑ cudaMalloc

Correct
Correct. This is not the correct function as cudaMalloc is used to allocate memory on the GPU, malloc is used in allocating CPU pageable memory.

- ☑ ::

Correct
Correct. This is used as part of C++'s namespacing convention but what is the correct replacement is the ; (semi-colon).

- ☐ sizeof
- ☐ char *
- ☑ input_a

Correct
Correct. input_a was never declared, so it cannot be assigned a pointer

7. Which of the following functions is used to allocate pinned CPU memory? **1 / 1 point**

- ◉ cudaMallocHost
- ○ print
- ○ malloc
- ○ cudaMallocManaged

Correct
This is correct as this functions purpose is to allocate memory for use by CUDA.

8. The purpose of struct allocation is to give a more object-orientted access pattern for host memory. **1 / 1 point**

- ◉ True
- ○ False

Correct
Structs are used to allocate objects that have named properties but do not have the functions that are used in higher-level classes in C++.

9. Given the following example device memory allocation: **1 / 1 point**

cudaError_t **cudaErr** = **cudaMalloc**(**&dev_a**,**N*sizeof(int)**);

Which of the following mappings of term to purpose is correct?

- ☑ N*sizeof(int) - This computes the size of an int array of size N

Correct
This is correct as it is a common practice to determine the size of the pointer to an array of integers.

- ☑ cudaErr - This is an instance of the data type used to hold error data

Correct
This is correct as the cudaError_t is a data type for holding error data.

- ☐ cudaMalloc - This is a function that allocates pageable GPU memory
- ☑ &dev_a - This is an int array pointer

Correct
This is correct as the & is used to indicate that the pointer will be passed by reference and the N*sizeof(int) indicates that the pointer is not to a single primitive but an array of ints.

10. Which of the following is a correct example of a copy of data from CPU pageable/pinned memory to GPU memory, presuming that h_a is host memory and d_a is device memory. **1 / 1 point**

- ○ cudaMemcpy(h_a, d_a, cudaMemcpyDeviceToHost);
- ◉ cudaMemcpy(h_a,d_a,cudaMemcpyHostToDevice);
- ○ cudaMemcpy(d_a,h_a,cudaMemcpyHostToDevice);
- ○ cudaHostGetDevicePointer((int **)&d_a, (int *)h_a, N*sizeof(int));

Correct
This is the correct way of copying data from host to device memory